

Model Formulation

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General Considerations on the formulation of phase field models

Phase field models are, quite generally, extensions of classical non-equilibrium thermodynamics. There are quite a few treatments of this in the literature. Here we will follow formulations similar to those developed by Sekerka and Bi in “Interfaces for the Twenty-First Century.” [\[BS98\]](#) In that spirit, we will start with a very general formulation, including hydrodynamics, and then simplify the problem. For those interested in starting with a simple model, you can skip over much of the initial formulation and jump to the section on the further reduction of the problem.

In thermodynamics one usually starts from consideration of the internal energy E , where $E = E(S, V, M_i)$, and S is the entropy, V is the volume and M_i is the mass of species i . If we decide we don't want to consider elasticity we can divide through by the volume and compute the bulk energy density $e = e(s, \rho_i)$, with $\rho_i = M_i/V$, and similarly $s = S/V$. It is then usually postulated that in addition to the thermodynamic variables there are any number of *phase fields* ϕ_j and the internal energy density is written

$$e = e(s, \rho_i, \phi_j).$$

Here we are going to consider a single phase field ϕ that distinguishes between two phases (could be liquid-solid, could be solid-gas, etc.) and postulate that the total entropy of a system of volume V can be written as

$$S = \int s^{NC} dV,$$

where

$$s^{NC} = s - \frac{\epsilon^2}{2} |\nabla \phi|^2 - \sum_i \frac{\alpha_i^2}{2} |\nabla \rho_i|^2.$$

Here the superscript NC is meant to indicate that this is a non-classical definition of the entropy density s . We have chosen the simplest possible forms of the non-classical extensions (that is, simple square-gradient penalties). The entropy density is itself a function, of course,

$$s = s(e, \rho_i, \phi).$$

To close the deal we now need to work our way through the various continuum-versions of the laws of nature.

Evolution equations

In order to write down evolution equations we need to resort to the known laws of physics. For continuum theories, we really have only a few ideas to fall back on

- Conservation of mass
- Conservation of energy (First Law of Thermodynamics)
- Conservation of momentum and maybe angular momentum if absolutely necessary
- The second law of thermodynamics (entropy increases, free energy decreases...)

A complete discussion of how to use the above rules in this context is outside the scope of this best-practice guide, but we offer a highly abbreviated discussion of the “flavor,” following the ideas of irreversible thermodynamics. For further reading, we recommend consulting works like deGroot and Mazur [[GM13](#)], although the continuum mechanics community may be more comfortable with Noll, Coleman, and Truesdale [[Mal69](#)] as well as the aforementioned work by Sekerka and Bi [[BS98](#)].

Mass

The first idea is that we need to specify a flux of particles

$$\mathbf{J}_i = \rho_i (\mathbf{v}_i - \mathbf{v}),$$

where $\mathbf{v}_i$ is the velocity of the particles of species i . In general we want to work in a center-of-mass frame

$$\mathbf{v} = \frac{\sum_i \rho_i \mathbf{v}_i}{\sum_i \rho_i}.$$

Notice that this implies (by construction) that $\sum_i \mathbf{J}_i = 0$.

The law of conservation of mass can be simply written as

$$\frac{\partial \rho_i}{\partial t} + \nabla \cdot (\rho_i \mathbf{v}_i) = 0.$$

where t is time. Then we use the definition of $\mathbf{J}_i$ to rewrite that as

$$\frac{\partial \rho_i}{\partial t} + \nabla \cdot (\mathbf{J}_i + \rho_i \mathbf{v}) = 0.$$

Note that if we sum this equation over i , we get the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0.$$

At this point it is common to introduce the notation

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla,$$

the “Lagrangian derivative” or “material derivative”. Then we can write the mass conservation equation as

$$\frac{D\rho_i}{Dt} + \nabla \cdot \mathbf{J}_i + \rho_i \nabla \cdot \mathbf{v} = 0.$$

If the system is incompressible (a common assumption, although we’d have to abandon that if we want to consider gasses), then $\nabla \cdot \mathbf{v} = 0$ and we can write

$$\frac{D\rho_i}{Dt} + \nabla \cdot \mathbf{J}_i = 0.$$

Momentum

The law of conservation of momentum (Newton's second Law) for a system without body forces (like gravity) can be written

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} - \sigma) = 0,$$

where σ is the stress tensor. Using continuity and incompressibility as assumptions we get

$$\rho \frac{D\mathbf{v}}{Dt} = \nabla \cdot \sigma.$$

Energy

Our final conservation expression is for energy. Just like the equation for momentum and mass, we start with a continuity form, and assume a flux of internal energy $\mathbf{J}_e$, and distinguish between the total energy density e_T , the internal energy density e and the kinetic energy density $e_k = e_T - e = \frac{1}{2} \rho v^2$ and write

$$\frac{\partial e_T}{\partial t} + \nabla \cdot (e_T \mathbf{v} + \mathbf{J}_e) = 0.$$

We can then use the momentum equation as well as the mass continuity equation (along with incompressibility) to arrive at

$$\frac{De}{Dt} + \nabla \cdot \mathbf{J}_e = \nabla \mathbf{v} : \sigma.$$

First Law of Thermodynamics

We now write down the laws of thermodynamics for the densities.

$$e = Ts - p + \sum_i \rho_i \mu_i,$$

where p is the hydrostatic pressure,

$$p = -\frac{1}{3} \text{Tr} \sigma,$$

and μ_i and T are defined as derivatives of e through the expression

$$de = Tds + \sum_i \mu_i d\rho_i + \frac{\partial e}{\partial \phi} d\phi.$$

Of course, given where we're headed, let's rewrite this as

$$ds = \frac{1}{T} de - \sum_i \frac{\mu_i}{T} d\rho_i + \frac{\partial s}{\partial \phi} d\phi.$$

Second Law

We have chosen the entropy to "extend" non-classically, as it is the star actor in the second law of thermodynamics, so we can write the entropy production (which must be positive) as

$$S^{\text{prod}} = \frac{dS}{dt} + \int \mathbf{J}_s \cdot d\mathbf{A} \geq 0,$$

where $\mathbf{J}_s$ is the flux of entropy. We can compute (for an incompressible system)

$$S^{\text{prod}} = \int \left[\sum_i -\left(\frac{\mu_i}{T}\right)^{NC} \frac{D\rho_i}{Dt} + S_\phi \frac{D\phi}{Dt} + \frac{1}{T} \frac{De}{Dt} \right] dV + \int \left[\mathbf{J}_s - \sum_i \alpha_i^2 \nabla \rho_i \frac{D\rho_i}{Dt} - \right]$$

where we have introduced

$$-\left(\frac{\mu_i}{T}\right)^{NC} = \frac{\partial s}{\partial \rho_i} + \alpha_i \nabla^2 \rho_i,$$

and

$$S_\phi = \frac{\partial s}{\partial \phi} + \epsilon^2 \nabla^2 \phi.$$

At this point we can insert the equations of motion we got from our conservation laws and do *a lot* of algebra, which we leave as an amusing exercise for the reader. We eventually find that

$$S^{\text{prod}} = \int \left[- \sum_i \nabla \left(\frac{\mu_i}{T} \right)^{NC} \cdot \mathbf{J}_i + S_\phi \frac{D\phi}{Dt} + \nabla \frac{1}{T} \cdot \mathbf{J}_e + \frac{1}{T} \mathbf{Y} : \nabla \mathbf{v} \right] dV,$$

and

$$\mathbf{J}_s = s^{NC} \mathbf{v} + \frac{1}{T} \mathbf{J}_e - \sum_i \left(\frac{\mu_i}{T} \right)^{NC} \mathbf{J}_i + \sum_i \alpha_i^2 \frac{D\rho_i}{Dt} \nabla \rho_i + \epsilon^2 \frac{D\phi}{Dt} \nabla \phi.$$

It is worth noting that this form for $\mathbf{J}_s$ eliminates the explicit surface terms from the entropy production. We have also introduced the tensor $\mathbf{Y}$ which is a complicated beast and has the form

$$\frac{\mathbf{Y}}{T} = \frac{\sigma}{T} + \sum \alpha_i^2 \nabla \rho_i \otimes \nabla \rho_i + \epsilon^2 \nabla \phi \otimes \nabla \phi + \left(\frac{p}{T} - \sum_i \alpha_i^2 \rho_i \nabla^2 \rho_i - \sum_i \frac{\alpha_i^2}{2} |\nabla \rho_i|^2 - \right.$$

where $\mathbf{I}$ is the identity tensor.

Constitutive Equations and Evolution Equations

After all this work we can now derive the constitutive equations. Inspection of the final expression for S^{prod} , suggests a that we can guarantee it be positive definite (which is required from the second law) by assuming

$$\mathbf{J}_i = -M_i \nabla \left(\frac{\mu_i}{T} \right)^{NC},$$

$$\mathbf{J}_e = M_T \nabla \frac{1}{T},$$

$$\mathbf{Y} = \eta (\nabla \mathbf{v} + (\nabla \mathbf{v})^T),$$

$$\frac{D\phi}{Dt} = M_\phi S_\phi,$$

where we have introduced mobilities and the viscosity η . It should be emphasized that we could have assumed all sorts of “cross-couplings” between the various fluxes here, but instead we have made the simplest assumptions. Inserting these into our equations of motion, we (at last!) arrive at general evolution equations

$$\frac{De}{Dt} = -M_T \nabla^2 \frac{1}{T} + \nabla \mathbf{v} : \sigma,$$

$$\frac{D\rho_i}{Dt} = \nabla \cdot M_i \nabla \left(-\frac{\partial s}{\partial \rho_i} - \alpha_i^2 \nabla^2 \rho_i \right)$$

$$\frac{D\phi}{Dt} = M_\phi \left(\frac{\partial s}{\partial \phi} + \epsilon^2 \nabla^2 \phi \right).$$

$$\frac{\sigma}{T} = \frac{\eta}{T} (\nabla \mathbf{v} + (\nabla \mathbf{v})^T) - \sum_i \alpha_i^2 \nabla \rho_i \otimes \nabla \rho_i + \epsilon^2 \nabla \phi \otimes \nabla \phi - \left(\frac{p}{T} - \sum \alpha_i^2 \rho_i \nabla^2 \rho_i - \sum \right)$$



with

$$\rho \frac{D\mathbf{v}}{Dt} = \nabla \cdot \sigma.$$

Further reduction of the problem

We have presented the derivation of equations of evolution for an incompressible, two-phase, multicomponent system. Most phase field treatments ignore the velocity terms, but for pedagogical purposes we have retained all of the terms that arise from flow (for an incompressible system). These terms should not just be tossed away without careful consideration of the system of interest. Nonetheless, now that we have taken care to show how these terms arise, and also, for careful readers, how to extend this approach to multiple phases and additional gradient corrections, we will proceed with some more simplifications for a less complex system. For those who are interested in solid state systems that can creep, the work of Mishin, Warren, Sekerka, and Boettinger (2013) [[MWSB13](#)] extends this framework. Here we eliminate the $\mathbf{v}$ equations by fiat, assuming that only diffusion controls

the evolution of the system, which is often reasonable in microgravity situations. We can also go further, and consider an isothermal system. Then we have

$$\frac{\partial \rho_i}{\partial t} = \nabla \cdot M_i \nabla \left(-\frac{\partial s}{\partial \rho_i} - \alpha_i^2 \nabla^2 \rho_i \right),$$

$$\frac{\partial \phi}{\partial t} = M_\phi \left(\frac{\partial s}{\partial \phi} + \epsilon^2 \nabla^2 \phi \right).$$

This system of equations should be adequate to describe a diffusion-controlled multicomponent, isothermal, two-phase system. One could, of course, retain thermal diffusion, add more phases, retain the velocity terms and more.

Clearly define the equations, parameters, initial conditions, and boundary conditions

Now that we have specified that we wish to consider a diffusion-controlled multicomponent, isothermal, two-phase system, we still need to decide on the specifics of the system. In particular, the thermodynamic state functions that detail how the energy or entropy vary. All of the variables need to be clearly defined, and ideally should be either parameters that can be traced to thermodynamic variables or other physical constraints. As noted, we wish to consider an isothermal system, and so we can use a Legendre transformation to replace s in the above expressions with the Helmholtz free energy $f = e - Ts$.

$$\frac{\partial \rho_i}{\partial t} = \nabla \cdot M_i \nabla \left(\frac{1}{T} \frac{\partial f}{\partial \rho_i} - \alpha_i^2 \nabla^2 \rho_i \right)$$

$$\frac{\partial \phi}{\partial t} = M_\phi \left(-\frac{1}{T} \frac{\partial f}{\partial \phi} + \epsilon^2 \nabla^2 \phi \right).$$

Additionally, we will now only consider only a two component systems, so we only have species ρ_1 and ρ_2 . Note that the total density $\rho = \rho_1 + \rho_2$ is a constant (for an incompressible system). We could formulate the problem in terms of these variables or we can move to a concentration $c_1 = \rho_1/(\rho_1 + \rho_2)$. Clearly $c_1 + c_2 = 1$, so we only have one independent variable. In this case we can pick c_1 or c_2 as the independent variable and label it c . As the reader can see, these types of exercises are highly non-trivial, require substantial care, and need to be done with rigor to understand the host of approximations and

assumptions that can be made (at a minimum here we have ignored elasticity, different molar volumes for the species, external fields, and there will be more to come). We move to a concentration picture and *redefine* the mobilities, α , and ϵ , in such a way that the following expressions hold:

$$\frac{\partial c}{\partial t} = \nabla \cdot M_i \nabla \left(\frac{\partial f}{\partial c} - \alpha^2 \nabla^2 c \right)$$

$$\frac{\partial \phi}{\partial t} = M_\phi \left(\epsilon^2 \nabla^2 \phi - \frac{\partial f}{\partial \phi} \right).$$

State function

We now need to specify $f(c, T, \phi)$. Note that we have retained the T dependence here, even though we are considering an isothermal system. Isothermal does not mean temperature independent. Additionally, in the expressions just above we “absorbed” T into some of the coefficients. These parameters can depend on T , but often the dependencies are unknown or should only be included if the phenomena under consideration demand such an inclusion. In the interest of pedagogy, we recapitulate one approach to modeling a liquid-solid binary alloy, although the details are less important than understanding that a specific choice of state function has to come from *somewhere*. Following the treatment in the Annual Reviews of Materials Research (2001) by Boettinger, Warren, Beckerman and Karma [\[BWBK02\]](#) we note that the free energy can be determined through a multi-step process where the two components are called A and B respectively:

Step 1

Start with the ordinary free energy of the pure components as liquid and solid phases, $f_L^A(T)$, $f_L^B(T)$, $f_S^A(T)$, $f_S^B(T)$. They are functions of temperature only.

Step 2

Form a function that represents both liquid and solid for pure A

$$f^A(\phi, T) = (1 - p(\phi))f_S^A + p(\phi)f_L^A + W^A g(\phi).$$

This function combines the free energies of the liquid and solid with the interpolating function $p(\phi)$ and adds an energy hump, $W^A g(\phi)$, between them. A similar expression can be obtained for component B. Note that $g(\phi)$ has minima at $\phi = 0$ and $\phi = 1$ with a maximum at $\phi = 0.5$ while $p(0) = 0$, $p(1) = 1$ and $p' = 0$ when $\phi = 0$ or $\phi = 1$. A popular choice for these functions is $p(\phi) = \phi^3(10 - 15\phi + 6\phi^2)$ and $g(\phi) = \phi^2(1 - \phi)^2$. Note also the $\partial p / \partial \phi = 30g(\phi)$ by construction.

Step 3

Form the function, $f(\phi, c, T)$, that represents (for example) a regular solution of A and B,

$$f(\phi, c, T) = (1 - c)f^A(\phi) + cf^B(\phi) + RT((1 - c)\ln(1 - c) + c\ln c) + c(1 - c)[\Omega_L(1 - \phi) + \Omega_S\phi]$$

where Ω_L and Ω_S are the regular solution parameters of the liquid and solid that again are combined with the interpolating function $p(\phi)$. The gas constant R needs to be chosen with the correct normalization so that the expression RT has units of energy per unit volume.

With this specific choice of a regular solution free energy we at last have fully specified a system and its evolution.

Clearly define and pose your physical problem before solving

Regardless of the explicit functional choices for the state function, the most important point made above is that the phase field method requires an explicit choice for this function. Once we have selected it, the equations derived above detail the evolution of the phase, composition, temperature, etc. Here we chose only to consider phase and composition, with temperature as a parameter. Having fully specified the state function, the next question is really about practical issues around a specific simulation. What are the boundary conditions? What are the relevant length scales? Here is a list of issues:

Complete dimensional analysis and understanding relevant scales; state the reference frame

There are any number of dimensionless parameters that can be found in these models. Understanding their relative role is beyond the scope of this best practice guide, but constructing these numbers is straightforward and essential. Usually before attempting to solve the equations, it is useful to "scale" the equations, thereby reducing the total number of

parameter in the problem. In general we can always rescale space and time to eliminate two parameters. For example, a natural unit for length ℓ might be defined in terms of a scale of the free energy density $\bar{f}$

$$\ell^2 = \frac{\epsilon^2}{\bar{f}}.$$

Using this definition of ℓ we can then get a time scale τ as

$$\tau = \frac{1}{M_\phi \bar{f}}.$$

Using these units will eliminate two parameters that need to be explored when seeking solutions.

Plot your state functions before implementing them in code

While we derived a specific choice of the free energy above, regardless of where you got the free energy (or other state function), it is wise to plot the function, as this will give you a quick check that your state function is well posed and has minima where you expect them to be.

Consider the impact of interpolation functions, barrier functions, etc.

We chose specific forms for our interpolation function p and the barrier function g . These were not arbitrary choices, but nonetheless, they are hardly the only choices we could have made, just relatively popular. All such choices can result in simulation regions where "surprises" can occur, often due to insufficient spatiotemporal resolution.

Consider every boundary term and make sure you aren't eliminating important terms. Check the fluxes

When implementing the boundary conditions, extreme care must be taken. One cannot just "zero-out" boundary terms, as this may break some requirement about mass conservation or other physical constraint.

Having clearly defined the mathematical framework underlying the problem, it becomes important, prior to progressing further, to contemplate and thoroughly grasp the various assumptions and approximations inherent within the formulation. Often, the exacting mathematical underpinnings from which the model has arisen may have presented challenges of being intractable in their complexity, or alternatively, they might have been of an impractical nature or even lacking contact with empirical observations.

The considerations surrounding assumptions and approximations are distributed throughout various places throughout the model's formulation. Foremost, are the descriptors of thermodynamic energy density, which provide the basis for our understanding of energy relationships, as well as the delineations of boundary conditions and the specifications of initial conditions. Notably, the latter two warrant attention since, in the absence of these conditions, the partial differential equations that determine the temporal evolution of our problem could only hypothetically yield an infinite set of solutions.

- If possible, run a small test problem with and without approximations to quantify their impact
- Consider how approximations change in different dimensions and symmetries
- If possible, avoid assuming symmetry to reduce the complexity of your problem

Carefully consider your length and time scales

1. **Identify the Smallest Feature Size of Interest in Your Problem:** The smallest feature size refers to the smallest spatial scale that needs to be accurately resolved in your simulation. This could be, for example, the width of an interface between two distinct materials or the size of a fine structure. Properly resolving the smallest features is essential for capturing detailed interactions and behaviors in your system.

Generally the smallest feature is the interface width between the features that need to be resolved. The interface width often represents the smallest detail that demands attention. Neglecting this width may lead to inaccurate results and incomplete understanding of the system's behavior.

2. **Identify the Smallest Time Scale of Interest in Your Problem:** The smallest time scale corresponds to the fastest processes occurring in your system. It's important to accurately capture these time scales to ensure that rapid events are not overlooked or misrepresented.

The time steps you consider shall be small enough to allow capturing the kinetics of the process you want to model. Choosing appropriate time steps is crucial to accurately capture dynamic processes. Smaller time steps are necessary to capture fast kinetics and prevent underestimation or distortion of time-dependent phenomena.

3. **Identify the Smallest Length Scales of Interest for All Physics and Consider How They Interact:** This involves analyzing the smallest spatial scales relevant to each physical process in your simulation. Understanding how these different length scales interact is essential for capturing multi-scale effects accurately.

Depending on the numerical method used for solving the equations, you may need 5-10 mesh elements to resolve the interface. The mesh resolution, i.e., the number of mesh elements used to discretize the domain, plays a crucial role in capturing interfaces accurately. Higher mesh resolution is required to resolve fine interfaces properly.

4. **The Sample Size You Consider Will Be Relevant to Your Feature Size:** The size of the simulation domain should be appropriately chosen based on the scale of the features you are interested in. Having a domain that is much larger than necessary can lead to unnecessary computational costs, while having a domain that is too small might exclude important interactions.
5. **It Is Recommended to Formulate Your Problem in a Non-Dimensional Form If Possible:** Non-dimensionalization involves scaling variables in your equations to remove physical units. This can help simplify the equations, reduce the number of parameters, and make the problem more amenable to analysis and comparison.
6. **If You Consider an Initial Condition Far from the Expected Steady State Solution, You May Need to Select a Significantly Smaller Time Step:** When the initial condition differs significantly from the expected steady state, smaller time steps may be necessary to accurately capture the transient behavior and prevent numerical instability.
7. **Be Mindful of Initial Conditions You Select:** Poorly chosen initial conditions, especially for interfaces or abrupt changes, can lead to numerical artifacts, excessive computational demands, and inaccurate results. It's important to select physically meaningful initial conditions that smoothly transition into the desired system state.
They may also result in numerical artifacts that lead to artificial stabilization of unstable phases. Inaccurate initial conditions can stabilize phases that would otherwise be unstable, leading to unrealistic outcomes in your simulation.
8. **Using Automatic Time Stepping Integrator Algorithms Can Benefit PF Simulations:** Automatic time stepping algorithms adjust the time step size during simulation based on the system's dynamics. This can improve simulation efficiency by using larger time steps during stable periods and smaller steps during dynamic events.

Summary

In summary, these considerations highlight the importance of careful parameter selection, proper resolution of length and time scales, and accurate initial conditions when performing numerical simulations. Addressing these aspects helps ensure the reliability and validity of your simulation results in various scientific and engineering applications.

- Non-dimensionalization

- It gets MUCH harder with coupled physics, so be careful

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